

Research of Adaptive Routing Scheme for LEO Network

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Abstract—Low Earth Orbit (LEO) satellite network is becoming an increasingly hot topic in satellite communication since it provides shorter round-trip delays compared with Geostationary Earth Orbit (GEO) satellites. Due to the rapid topology changes in the LEO constellation, its routing protocol should be different from existing protocols harnessed for the terrestrial networks. Based on the periodicity and predictability characteristics of LEO network, static routing algorithms are most commonly used. The topology snapshots and corresponding forwarding tables are calculated in advance so as to save the on-board computing resources. However, the static routing algorithm is not compatible with sudden link failure, which reduces its robustness. In this paper, a static-dynamic combined routing strategy is proposed as an adaptive routing scheme to LEO satellite networks. The new scheme captures both the higher robustness compared with static routing algorithms and lower packet loss, higher throughput compared with dynamic routing algorithms.

Keywords—LEO; constellation topology; static routing algorithm; dynamic routing algorithm

I. INTRODUCTION

Recently, Low Earth Orbit (LEO) network is becoming a trendy solution in global networking. To construct a satellite network, the routing strategy is the first issue to consider [1]. Compared with Geostationary Earth Orbit (GEO) and Medium Earth Orbit (MEO) satellites, LEO satellites are able to provide connections with lower latency and higher speed [2]. However, the topology of LEO constellation also changes more frequently [3]. When using dynamic routing protocol such as OSPF (Open Short Path First) [4], which is commonly used in terrestrial networks, the routing calculation overheads will considerably occupy the limited on-board computing resources. Moreover, due to the long link delay and large constellation scale, dynamic routing protocols are difficult to achieve a fast convergence after topology changes, resulting in an increase in packet loss rate and a decrease in throughput [5].

For LEO satellites, the constellation orbit and satellite arrangement are determined and the surrounding period of each satellite is usually around 100 minutes, so the topology changes are predictable and can be abstracted into a snapshot sequence. This is one of the static routing protocols commonly used in satellite networks [6]. The basic idea of static routing is to calculate the network topologies in advance. By dividing the constellation period into several

time slices, the network topology in each time slice can be regarded as a virtual fixed topology. According to these topology snapshots, their corresponding routing and forwarding tables can be calculated and bounded on satellites, saving flooding overhead and computing cost on-board. When a data packet arrives, the only task of the on-board router is to look up the table and forward it. The static routing problem in satellite networks has been studied extensively in the past. ATM routing protocol in [7], FSA-based routing protocol in [8] and the snapshot-based routing protocol in [9] can all be classified as system-cycle segmentation algorithms. Static routing pre-computes an optimized path for each source-destination pair in advance [10]. However, the drawback of this mechanism is that it cannot adapt to unexpected link failures and network structure changes. When a node failure or congestion happens, continuing to route the packets according to the original topology may result in packet loss, which may cause repeated retransmission and results in heavier congestion situation in return.

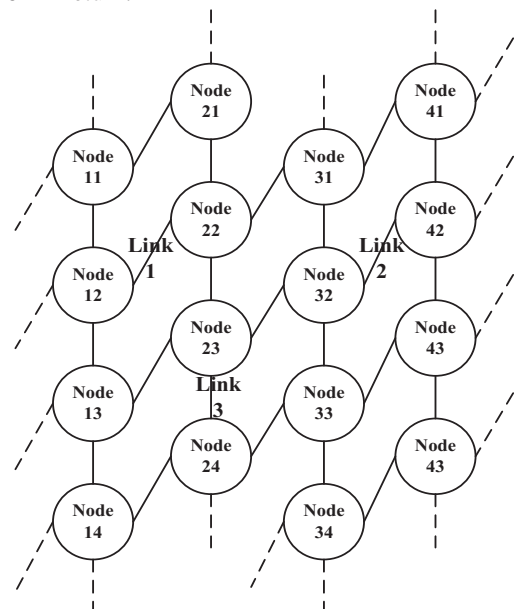


Figure 1. Structure of mesh network topology

In a typical LEO satellites model, each satellite can be seen as a node in the network. Each node has four ports, two of them connected to two nearest nodes in adjacent orbits

respectively and the rest of them connected to two neighbor satellites in the same orbit. As a typical undirected mesh network, part of the network topology can be abstracted into the structure shown in Fig. 1. In this paper, inter-satellite link refers to links between satellites in same orbits (Link 3 in Fig. 1). Intra-satellite link refers to links between adjacent orbits (Link 1 in Fig. 1). Inter Satellite Link (ISL) is the general name to the overall links in satellite networks.

By combining and switching between static and dynamic routing protocol, the new routing strategy is fault tolerant to unexpected topology changes while being able to maintain a lightweight mode in normal states. In this paper, we describe a more precise method specifically to generate snapshot series. By doing so, the strategy can control the packet loss during snapshot switching to a relatively low value. Different from terrestrial facilities, recovery of on-board equipment can be slow and time-cost. To avoid being stuck in dynamic routing when stubborn failure occurs, the strategy employs a temporary static routing mode in this situation. In this way, our strategy captures both lower packet loss and control packet overhead compared to OSPF, thus increasing throughput. When compared to static routing algorithm, our new strategy has a higher robustness when facing unexpected node or link failures.

This paper is organized as follows. In Sect. 2, we presents our routing strategy in detail. The simulation result and comparison with OSPF is described in Sect. 3. Finally, Sect. 4 concludes the paper.

II. STATIC-DYNAMIC COMBINED STRATEGY

The adaptive routing scheme proposed in this paper is a combination of snapshot-based static routing protocol and modified-OSPF dynamic protocol. It employs the link detection and broadcast mechanism to recognize whether the current network topology matches the snapshot. When the actual network topology matches the snapshot, the router only performs the operation of looking up the forwarding table and forwarding the data packet. When the two don't match, the routing table is recalculated continuously on each node in the entire network, and dynamic routing is maintained until the network topology matches the static snapshot again. When the network is in the static routing phase, snapshots are switched in time series. During a snapshot period, the routing table is pre-calculated based on the network topology reflected by the snapshot. In this section, we also propose a generating and partitioning strategy for snapshot sequence in order to prevent the snapshot switching from being too frequent and the packet loss caused by the switching process.

A. Link Detection and Broadcast Mechanism

Satellite routers periodically send HELLO packets between short intervals to detect available neighbor routers. When a port of the router receives HELLO from any other router, the host router will recognize if it is from its neighbor routers. If so, the bidirectional link will be established as long as the host ID also appears in the response packet from one of its neighbor routers. Otherwise, the HELLO packet will be discarded. In addition, the HELLO session is also

responsible for the maintenance of the connection relationship. The link will be determined disconnected when the last received HELLO packet from the neighbor router exceeds the dead time interval.

As the basis for the link detection, the HELLO session should always run in a high frequency. To ensure that each satellite router in the constellation maintains the same topology, which is the link state database (LSDB) accordingly for each router, link state announcements (LSA) are required to be broadcasted among the constellation. When the link state of a certain router changes or when it reaches the refresh time, the router sends an LSA packet to all of its neighbors. The flooding method in this session so as to ensure each router in the network can get a copy of the status change information. When other satellites receive the latest LSA, they will first compare the link state information to its own LSDB. If different, the database will be updated. Each satellite then calculates the routing table according to its LSDB. Regard itself as the root node, each satellite has to calculate the optimal paths to all the other routers in the network and store them to the routing table. This process is quite similar to OSPF in terrestrial network essentially.

In the static routing phase, a lower-frequency LSA refresh policy can be set. It can be performed only when the HELLO session detects the link state change, so as to reduce the signaling overhead in the static routing phase. In contrast, each satellite node needs to perform LSA flooding at a higher frequency in the dynamic routing phase to make sure that link changes can be quickly learned by the entire network.

B. Static Routing Scheme

When there is no link or node failure in the system, the snapshot only switches at time $t_i (i \in T)$ and remains unchanged during the time slot $[t_i, t_{i+1}]$. Packets routing in this interval follow the i^{th} topology snapshot and its corresponding routing table. In this single time slice, packet routing can be seen as a static topology route, which effectively simplifies the problem. In this static scheme, routing table is calculated using Dijkstra algorithm, and packets sent in the $[t_i, t_{i+1}]$ interval are forwarded according to the routing table calculated in advance. To minimize the packet drop during snapshot switching, we propose the following method for snapshots generating.

When the connection relationships among satellites are the same as the current time snapshot, it is considered that there's no link failure or node breakdown in the network. The static routing scheme is adopted by the network in this case. Although the network topology is changing from time to time due to satellite movements, it remains unchanged in each certain time period and is repeated in every constellation cycle. Based on the periodic characteristic of the satellite constellation, the static routing scheme cut the system cycle into slices by simulating the running state of the constellation and generate discrete finite number of topological snapshots. In a time set $T = \{t_0, t_1, \dots, t_N\}$, each element represents time point when the topology changes. The switching point is based on the change of a single inter satellite link (ISL). The network topology can be represented

as a time varying sub-graph $G_i = (V_i, E_i)$ within a single time interval $[t_i, t_{i+1}]$. According to the time varying sub-graph, the topology $S_i = \langle t_i, G_i \rangle$ in this time interval can be obtained. In the full cycle, topologies of the network can be summarized as a snapshot sequence $S = \langle S_0, S_1, S_2, \dots \rangle$.

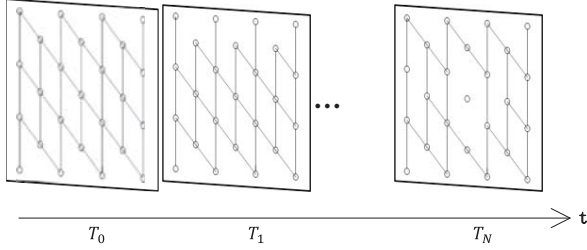


Figure 2. Discrete topology sequence

In the k^{th} snapshot period, the link cost between node i and j is represented by $c_{ij}^k(t)$, in which $t \in [t_k, t_{k+1}]$. The snapshot dividing rule should be as follows,

$$(c_{ij}^k(t) - c_{ij}^k(t_k)) / c_{ij}^k(t_k) \ll 1 \quad (1)$$

However in common constellation configurations, inter-satellite links at the same latitude (e.g. Link 1 and Link 2 in Fig. 1) usually switch off in a small certain period of time. For example, the relative speed of satellites in adjacent orbits is higher above circumpolar altitude. Links between adjacent orbits in these areas are usually not detected in most constellations. When several inter-satellite links arrive at a specified latitude, they are forced to be switched off almost simultaneously. If snapshots are still divided by single link change under these circumstances, they may be switched multiple times in a short period. To avoid being so, several topologies with small intervals should be merged. In Fig. 3, t_k represents the lifetime of k^{th} snapshot. During t_0 to t_8 , only one inter satellite link changes between every two contiguous snapshots. If $\sum_i t_i$ is too small and there are no link changes iteratively, which means each link changes only once, then the snapshots during this period can be merged. Two snapshots at t_0 and t_8 are reserved as snapshots N and $N+2$. Snapshots during t_1 to t_7 are merged into a transition state. The transition state is designed based on the N^{th} and $N+2^{th}$ snapshots. To avoid packet loss caused by snapshot merging and switching, changing links between these two snapshots are deemed to be disconnected in transition snapshot. Take $S[N]$ as the overall connected links in the N^{th} snapshot, the above rule can be expressed as follows,

$$S[N+1] = S[N] \cap S[N+2] \quad (2)$$

In addition, the start point of transition snapshot $N+1$ should be moved forward by δ_a time to enhance the reliability of the snapshot switching process. Also it should be terminated δ_b time prolonged. Take Fig. 3 as an example, suppose snapshots N and $N+1$ switches at time t_p . If a packet is sent out through a changing route at time $t_p - \Delta\tau$ (in which $\Delta\tau \ll t_0$) and doesn't get to its destination at t_p , it will be lost. The value of δ_a and δ_b can be modified according to the maximum transmission delay of packets in the network.

For a constellation with scale of $M \times N$ satellites (N orbital planes with M satellites on each plane), the network radius R refers to the maximum distance (hop count) that packet could be transmitted in the network,

$$R = \left\lceil \frac{M}{2} \right\rceil + \left\lceil \frac{N}{2} \right\rceil \quad (3)$$

The corresponding end-to-end delay should be as follows,

$$DT_{max} = \sum_{i,j} D_{ISL_{ij}} + R \cdot T_s \approx R \cdot (D_{ISL_{average}} + T_s) \quad (4)$$

Where $D_{ISL_{ij}}$ refers to the propagation delay of the packet between satellite i and j , which can be approximated by its average value $D_{ISL_{average}}$. T_s refers to the packet processing delay in each satellite router. To minimize the packet loss caused during the topology switching process, we suppose $\delta_a = \delta_b = DT_{max}$.

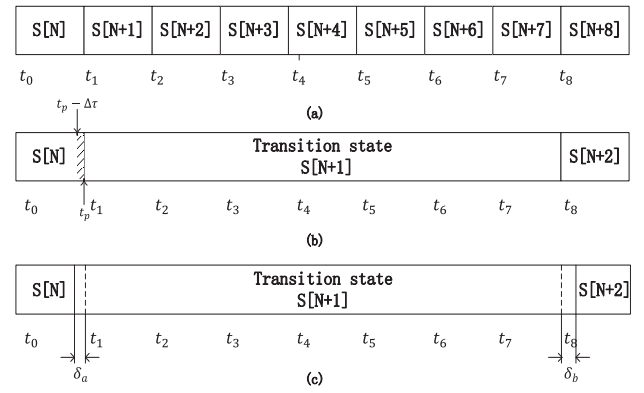


Figure 3. Example of snapshot merging

Topology snapshots can be pre-stored in on-board non-volatile memory or be sent to satellite by terrestrial stations in order to save on-board storage resources. The corresponding routing tables can also be pre-calculated and pre-stored. To save on-board memory resources, they can also be calculated onboard ahead of snapshots switching time.

C. Dynamic Routing Scheme

Dynamic routing scheme mainly holds the promise for the system performance when link or node failure occurs. In this case, the actual link state does not match the topology snapshot. Using static routing scheme continually may result in packet loss. Dynamic routing relies on HELLO and LSA session for link state detection and broadcast. In dynamic routing scheme, we employ shortest path algorithm Dijkstra to dynamically calculate the routing table in time. Based on the more frequently flooded LSA in the network compared to static routing scheme, each satellite is able to update LSDB by current link states and hold a same topology abstract synchronously. Therefore in regard to LSA flooding frequency, it should be increased in dynamic scheme to ensure the quick synchronization of link state changing information among the network.

D. Adaptive Static-Dynamic Switching Strategy

The Finite State Machine (FSM) of the adaptive routing scheme is illustrated in Fig. 3,

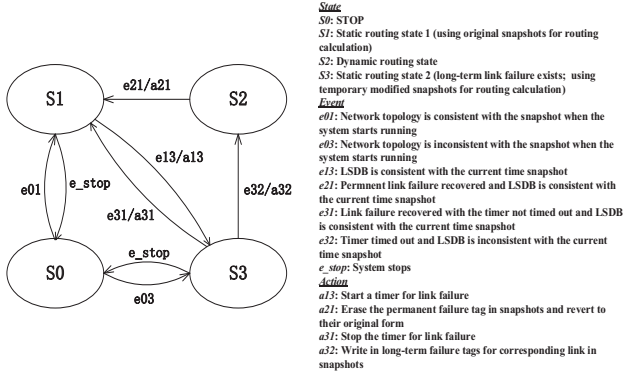


Figure 4. FSM for routing schemes

Link State Data Base is updating from time to time by HELLO and LSA session. Each time when LSDB changes, the system compares it with the current time snapshot. Static routing scheme keeps running when LSDB conforms to the corresponding snapshot topology since there's no link or node failure or they are supposed to be switched off during this snapshot period. When at least one link failure is detected and flooded by HELLO and LSA session respectively, each node in the network will switch into dynamic routing scheme because of the inconformity of the actual constellation topology and its corresponding snapshot. When HELLO session detects that the link failure is restored, the system will be switched back to the static routing scheme after LSA session broadcasts to the entire network and the LSDB is exactly the same with the current time snapshot.

The dynamic routing scheme calculates the routing table based on the current LSDB which has link failure in it. Whenever LSDB changes, the routing table needs to be recalculated. If the dynamic routing sustains when it takes a long time for broken link to rebuild connection, large on-board computing resources will be occupied and a huge amount of signaling overhead will be generated which will result in congestion and packet loss. To avoid getting stuck in dynamic routing scheme because of a certain link failure, the long term link failure should be tagged in the overall snapshots. Specifically, when the duration of node A unable to contact node B through HELLO session exceeds a threshold, this long-term link failure should be tagged in the full period snapshots and thus turning them into a temporary snapshot sequence. The system then switches to static routing scheme using new routing tables calculated according to the temporary snapshot sequence. When link failure recovers and there's no more link failure in the network, the system will recall the original snapshot sequence. Routers then forwards packets based on the original routing tables.

III. SIMULATION RESULT

In this section, we simulate and compare OSPF (Open Short Path First) which is commonly used in terrestrial network with our proposed adaptive routing scheme under the same satellite network and traffic configuration.

A. Network Configuration

The LEO network topology used in this context is similar to the Iridium satellite network [11]. The specific parameters are as follows.

TABLE I. NETWORK CONFIGURATION

Orbit altitude	1100km
Number of orbit planes	6
Number of satellite in each orbit	9
Orbit inclination	85°

Each satellite has four links with four neighbor satellites. Two link-connected satellites (shown in Fig. 1 as node 11 and 21) on adjacent orbits have a latitude interval of 30°. We stipulate that there is no link connection in the reverse seam which is shown in Fig. 5 as the shadow block. Since satellites in the polar area are too close to each other, it is also assumed that inter-satellite links are disconnected when satellite moves to the area with latitude higher than 60°, which means only intra-satellite links are needed to be detected maintained under these circumstances.

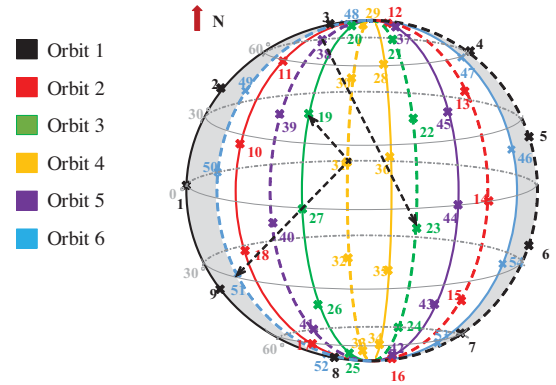


Figure 5. Satellite constellation

(Satellite on real line orbits move towards North; Satellite on imaginary line orbits move towards South)

After simulate the periodical characteristic of this satellite constellation, we merge the topology snapshot and get 18 snapshots left, each of them lasts for about 350s. Then we add 18 transition snapshots according to the method in Sect.2 B.

As shown in Fig. 5, three data flows are set into the network,

- Node 31 to Node 19;
- Node 38 to Node 34;
- Node 31 to Node 51.

The simulation time is 20 min. To ensure that the network system reaches the convergence state at its

beginning, data flows are loaded from 10s to the end of the simulation.

Data flow 2 and 3 pass Node 40 and 49 respectively when they start. To verify the robustness of the network, Node 40 and 49 are preset to be failed from 300s to 500s. In doing so, the adaptive ability can be tested.

B. Simulation Results

The throughput of a single data flow can be calculated according to the service data bits it received per unit time,

$$\text{throughput} = \frac{\sum_{i=1}^N L_i}{\Delta T} \quad (5)$$

Where N refers to the total number of data packets received in ΔT , L refers to the number of service data bits in each data packet. We set L_i to 10kbit in the simulation and change the sending rate by modifying the packet transmission interval.

The simulation result is shown in Fig. 6

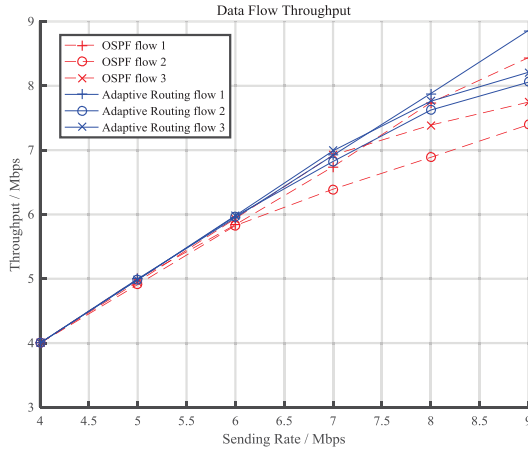


Figure 6. Simulation result of throughput

Next, we compare the total packet numbers of signaling overheads for OSPF and Adaptive Routing Scheme. When HELLO interval is set to be the same, the LSA overhead percentage of two method have an obvious difference as is illustrated in Table 2,

TABLE II. TOTAL NUMBERS OF LSA OVERHEAD

Throughput	OSPF	Adaptive Routing Scheme
4Mbps	0.41%	0.30%
5Mbps	0.32%	0.24%
6Mbps	0.26%	0.19%
7Mbps	0.23%	0.17%
8Mbps	0.20%	0.15%
9Mbps	0.18%	0.13%

Fig. 6 shows a roughly same performance of OSPF and Adaptive Routing when the traffic rate is low. However, the packet loss rate of OSPF increases significantly when the traffic rate is increased to 8 Mbps and thus pulling the throughput down. But Adaptive Routing Scheme performs better than OSPF when the traffic rate is higher than 8 Mbps. Its packet loss rate is much lower than that of OSPF. For data flow 2 and 3 which each has link failure occurs during the simulation, the throughput decreases of both OSPF and

Adaptive Routing Scheme. While comparing to static protocols, our routing scheme can effectively adapt to sudden link failures and control the packet loss rate to a certain range.

During the 20 min simulation period, the network goes through several topology changes. When there's no sudden link failure, the Adaptive Routing Scheme updates its routing table according to predicting topology changes, thereby decreasing the packet loss rate and maintaining a rather high throughput. It can be deduced from the simulation result that OSPF has difficulties adapting the highly dynamic topology changes of the LEO network. The require for in-time routing table update generates a large amount of signaling overhead. When the traffic volume increases, a large amount of packet loss will occur. It's concluded from simulation result that our Adaptive Routing Scheme is more suitable for the LEO network than OSPF when there are large amounts of traffic in the network.

IV. CONCLUSION

In this paper, we proposed an adaptive routing scheme for LEO networks. Considering the highly dynamic topology changes of the LEO network and the difficulty to restoring sudden link/node failures in a short time, we suggested combining dynamic routing with static routing which is commonly used in LEO networks.

We have carried out theoretical analysis and simulation results in the above sections and the results show that when comparing to dynamic routing methods, the Adaptive Routing Scheme is able to adapt to satellite networks with highly dynamic topology changes with less control signaling overhead, higher throughput and lower control signaling overhead. Compared with static routing algorithms commonly used in satellite networks, our proposed routing scheme can adapt to sudden link/node failures and reduce the packet loss caused by them. The methods of partitioning and switching snapshots are designed to reduce the packet loss during snapshot switches and improve robustness of the system.

ACKNOWLEDGMENT

This work was supported by a grant from the National Natural Science Foundation of China (No. 61972398).

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